

Replay-Aware TSP Benchmark Report

Overview

- Condition count: 9.
- Best final transfer gap: phase6_residual_failure_replay.
- Best TSPLIB holdout gap: phase6_residual_failure_replay.
- Best synthetic holdout gap: phase6_random_replay.

Run Metadata

- run_name: run_20260519_001148_f
- started_at_local: 2026-05-19 00:11:48
- finished_at_local: 2026-05-19 00:31:50
- duration_hhmm: 00:20
- duration_seconds: 1202.299
- seed_offset: 5000
- replicate_label: f
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_no_replay	none	score_only	False	0.232246	0.020958	0.155414	0.236772	0.170476	0.0	0.805346	0.76	-0.022765
phase6_random_replay	random	score_only	False	0.122006	0.0	0.07764	0.236772	0.145007	0.19457	0.685363	0.76	0.054841
phase6_failure_replay	failure	score_only	False	0.213387	0.064916	0.159398	0.171206	0.348017	0.261637	0.79099	0.56	-0.031486

Condition	Replay	Selection	Compression	Final TSPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Archive Diversity	Archive Hardness	Failure Concentration	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
phase6_random_replay_compression	random	novelty_gate	True	0.098726	0.0	0.062826	0.236772	0.113866	0.19457	0.817438	0.76	0.011577
phase6_stratified_random_replay	stratified_random	score_only	False	0.121156	0.010343	0.08086	0.236772	0.149538	0.221162	0.0	0.76	0.0
phase6_diversity_weighted_replay	diversity_weighted	score_only	False	0.13595	0.0	0.086514	0.236772	0.155695	0.130039	0.827986	0.76	0.023865
phase6_residual_failure_replay	residual_failure	score_only	False	0.082341	0.0	0.052399	0.129767	0.296745	0.276871	0.832596	0.76	0.039093
phase6_diversity_failure_replay	diversity_failure	score_only	False	0.096401	0.0	0.061346	0.171206	0.291624	0.17024	0.828815	0.76	0.032532
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	True	0.234344	0.022049	0.157146	0.171206	0.261064	0.17767	0.0	0.76	0.0

Condition Notes

phase6_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB 0.232246, synthetic 0.020958, combined 0.155414.
- Accepted-epoch count 2, mean accepted code novelty 0.805346, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.170476, size bias 8.3e-05, and failure concentration 0.0.
- Replay selection diversity 0.0, mean selected expected gap 0.0, mean selected residual gap 0.0.
- Adaptation efficiency -0.022765 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.232246 across 7 instances; family means: ch=0.241182, kroD=0.218371, pcb=0.275671, pr=0.322044, rd=0.268015, st=0.059259.
- Panel synthetic_holdout mean gap 0.020958 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.083832, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3388, "expected_gap": 0.406514, "family": "a", "name": "a280", "optimality_gap": 0.313687, "residual_gap": -0.092827}, {"best_known_cost": 7542, "cost": 9109, "expected_gap": 0.245107, "family": "berlin", "name": "berlin52", "optimality_gap": 0.20777, "residual_gap":

-0.037337}, {"best_known_cost": 14379, "cost": 17073, "expected_gap": 0.328729, "family": "lin", "name": "lin105", "optimality_gap": 0.187357, "residual_gap": -0.141372}]

phase6_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.122006, synthetic 0.0, combined 0.07764.
- Accepted-epoch count 2, mean accepted code novelty 0.685363, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.145007, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.13412, mean selected residual gap -0.025781.
- Adaptation efficiency 0.054841 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 2, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.122006 across 7 instances; family means: ch=0.129981, kroD=0.037757, pcb=0.201997, pr=0.15872, rd=0.170417, st=0.025185.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3529, "expected_gap": 0.406204, "family": "a", "name": "a280", "optimality_gap": 0.36836, "residual_gap": -0.037844}, {"best_known_cost": 629, "cost": 774, "expected_gap": 0.25628, "family": "eil", "name": "eil101", "optimality_gap": 0.230525, "residual_gap": -0.025755}, {"best_known_cost": 14379, "cost": 17377, "expected_gap": 0.328131, "family": "lin", "name": "lin105", "optimality_gap": 0.208499, "residual_gap": -0.119632}]

phase6_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.213387, synthetic 0.064916, combined 0.159398.
- Accepted-epoch count 3, mean accepted code novelty 0.79099, and final complexity 0.56.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.348017, size bias 0.120322, and failure concentration 0.261637.
- Replay selection diversity 0.177669, mean selected expected gap 0.310218, mean selected residual gap 0.065539.
- Adaptation efficiency -0.031486 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.213387 across 7 instances; family means: ch=0.1282, kroD=0.262046, pcb=0.264938, pr=0.350299, rd=0.22225, st=0.137778.
- Panel synthetic_holdout mean gap 0.064916 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.259662, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3614, "expected_gap": 0.408608, "family": "a", "name": "a280", "optimality_gap": 0.401318, "residual_gap": -0.00729}, {"best_known_cost": 629, "cost": 875, "expected_gap": 0.25469, "family": "eil", "name": "eil101", "optimality_gap": 0.391097, "residual_gap": 0.136407}, {"best_known_cost": 14379, "cost": 19107, "expected_gap": 0.329634, "family": "lin", "name": "lin105", "optimality_gap": 0.328813, "residual_gap": -0.000821}]

phase6_random_replay_compression

- Replay mode: random with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.098726, synthetic 0.0, combined 0.062826.

- Accepted-epoch count 3, mean accepted code novelty 0.817438, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.113866, size bias 8.3e-05, and failure concentration 0.19457.
- Replay selection diversity 0.262174, mean selected expected gap 0.132632, mean selected residual gap -0.059201.
- Adaptation efficiency 0.011577 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 1, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.098726 across 7 instances; family means: ch=0.122658, kroD=0.069597, pcb=0.196069, pr=0.085273, rd=0.074083, st=0.020741.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 7542, "cost": 10132, "expected_gap": 0.240918, "family": "berlin", "name": "berlin52", "optimality_gap": 0.34341, "residual_gap": 0.102492}, {"best_known_cost": 2579, "cost": 3407, "expected_gap": 0.409771, "family": "a", "name": "a280", "optimality_gap": 0.321055, "residual_gap": -0.088716}, {"best_known_cost": 14379, "cost": 17920, "expected_gap": 0.322762, "family": "lin", "name": "lin105", "optimality_gap": 0.246262, "residual_gap": -0.0765}]

phase6_stratified_random_replay

- Replay mode: stratified_random with selection mode score_only.
- Final transfer gaps: TSPLIB 0.121156, synthetic 0.010343, combined 0.08086.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.149538, size bias 8.3e-05, and failure concentration 0.221162.
- Replay selection diversity 0.187367, mean selected expected gap 0.236305, mean selected residual gap -0.048527.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.121156 across 7 instances; family means: ch=0.105887, kroD=0.111017, pcb=0.236067, pr=0.092327, rd=0.105057, st=0.091852.
- Panel synthetic_holdout mean gap 0.010343 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.041372, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3452, "expected_gap": 0.404343, "family": "a", "name": "a280", "optimality_gap": 0.338503, "residual_gap": -0.06584}, {"best_known_cost": 14379, "cost": 18103, "expected_gap": 0.322164, "family": "lin", "name": "lin105", "optimality_gap": 0.258989, "residual_gap": -0.063175}, {"best_known_cost": 7542, "cost": 9348, "expected_gap": 0.236622, "family": "berlin", "name": "berlin52", "optimality_gap": 0.239459, "residual_gap": 0.002837}]

phase6_diversity_weighted_replay

- Replay mode: diversity_weighted with selection mode score_only.
- Final transfer gaps: TSPLIB 0.13595, synthetic 0.0, combined 0.086514.
- Accepted-epoch count 2, mean accepted code novelty 0.827986, and final complexity 0.76.
- Active replay archive experience_archive with mean diversity 0.236772, mean hardness 0.155695, size bias 8.3e-05, and failure concentration 0.130039.
- Replay selection diversity 0.261003, mean selected expected gap 0.167655, mean selected residual gap -0.021397.

- Adaptation efficiency 0.023865 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.13595 across 7 instances; family means: ch=0.109508, kroD=0.201277, pcb=0.226771, pr=0.095766, rd=0.149558, st=0.059259.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3698, "expected_gap": 0.40318, "family": "a", "name": "a280", "optimality_gap": 0.433889, "residual_gap": 0.030709}, {"best_known_cost": 629, "cost": 733, "expected_gap": 0.254054, "family": "eil", "name": "eil101", "optimality_gap": 0.165342, "residual_gap": -0.088712}, {"best_known_cost": 14379, "cost": 15330, "expected_gap": 0.329244, "family": "lin", "name": "lin105", "optimality_gap": 0.066138, "residual_gap": -0.263106}]

phase6_residual_failure_replay

- Replay mode: residual_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.082341, synthetic 0.0, combined 0.052399.
- Accepted-epoch count 3, mean accepted code novelty 0.832596, and final complexity 0.76.
- Active replay archive residual_archive with mean diversity 0.129767, mean hardness 0.296745, size bias 0.069117, and failure concentration 0.276871.
- Replay selection diversity 0.148354, mean selected expected gap 0.277755, mean selected residual gap 0.061728.
- Adaptation efficiency 0.039093 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 3.
- Panel heldout_tsplib mean gap 0.082341 across 7 instances; family means: ch=0.062421, kroD=0.076453, pcb=0.217909, pr=0.038767, rd=0.042857, st=0.075556.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 2579, "cost": 3304, "expected_gap": 0.402947, "family": "a", "name": "a280", "optimality_gap": 0.281117, "residual_gap": -0.12183}, {"best_known_cost": 629, "cost": 726, "expected_gap": 0.262003, "family": "eil", "name": "eil101", "optimality_gap": 0.154213, "residual_gap": -0.10779}, {"best_known_cost": 7542, "cost": 7964, "expected_gap": 0.237364, "family": "berlin", "name": "berlin52", "optimality_gap": 0.055953, "residual_gap": -0.181411}]

phase6_diversity_failure_replay

- Replay mode: diversity_failure with selection mode score_only.
- Final transfer gaps: TSPLIB 0.096401, synthetic 0.0, combined 0.061346.
- Accepted-epoch count 5, mean accepted code novelty 0.828815, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.291624, size bias 0.120322, and failure concentration 0.17024.
- Replay selection diversity 0.199284, mean selected expected gap 0.300697, mean selected residual gap -0.009641.
- Adaptation efficiency 0.032532 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 4, 'adversarial_layouts': 4} / elite 5.
- Panel heldout_tsplib mean gap 0.096401 across 7 instances; family means: ch=0.06478, kroD=0.060862, pcb=0.218796, pr=0.120452, rd=0.087358, st=0.057778.
- Panel synthetic_holdout mean gap 0.0 across 4 instances; family means: clustered_gaussian=0.0, grid_outliers=0.0, ring_bridge=0.0, two_corridors=0.0.

- Worst recent training cases: [{"best_known_cost": 14379, "cost": 19073, "expected_gap": 0.328006, "family": "lin", "name": "lin105", "optimality_gap": 0.326448, "residual_gap": -0.001558}, {"best_known_cost": 2579, "cost": 3212, "expected_gap": 0.402171, "family": "a", "name": "a280", "optimality_gap": 0.245444, "residual_gap": -0.156727}, {"best_known_cost": 629, "cost": 714, "expected_gap": 0.267091, "family": "eil", "name": "eil101", "optimality_gap": 0.135135, "residual_gap": -0.131956}]

phase6_diversity_failure_replay_compression

- Replay mode: diversity_failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB 0.234344, synthetic 0.022049, combined 0.157146.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.76.
- Active replay archive worst_archive with mean diversity 0.171206, mean hardness 0.261064, size bias 0.120322, and failure concentration 0.17767.
- Replay selection diversity 0.201519, mean selected expected gap 0.315683, mean selected residual gap -0.035024.
- Adaptation efficiency 0.0 and archive sizes {'experience_cases': 10, 'worst_cases': 6, 'failure_cases': 6, 'residual_failures': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.234344 across 7 instances; family means: ch=0.194363, kroD=0.272518, pcb=0.236086, pr=0.336884, rd=0.26397, st=0.142222.
- Panel synthetic_holdout mean gap 0.022049 across 4 instances; family means: clustered_gaussian=0.010624, grid_outliers=0.077572, ring_bridge=0.0, two_corridors=0.0.
- Worst recent training cases: [{"best_known_cost": 629, "cost": 819, "expected_gap": 0.263911, "family": "eil", "name": "eil101", "optimality_gap": 0.302067, "residual_gap": 0.038156}, {"best_known_cost": 2579, "cost": 3262, "expected_gap": 0.402171, "family": "a", "name": "a280", "optimality_gap": 0.264831, "residual_gap": -0.13734}, {"best_known_cost": 14379, "cost": 15975, "expected_gap": 0.326657, "family": "lin", "name": "lin105", "optimality_gap": 0.110995, "residual_gap": -0.215662}]

Judge Appendix

Summary of Main Evidence (Held-out TSPLIB Gap & Synthetic Holdout Gap)

Condition	Replay Mode	Selection Mode	Archive Size	Final TSP LIB Gap	Final Synthetic Gap	Final Transfer Gap	Adaptation Efficiency	Code Novelty (Mean)	Complexity	Failure Concentration (Mean)	Replay Failure Concentration (Final)
phase6_no_replay	none	score_only	29	0.232246	0.020958	0.155414	-0.022765	0.805346	0.76	0.0	0.0
phase6_random_replay	random	score_only	28	0.122006	0.0	0.07764	0.054841	0.685363	0.76	0.19457	0.16
phase6_failure_replay	failure	score_only	30	0.213387	0.064916	0.159398	-0.031486	0.79099	0.56	0.261637	0.306122
phase6_random_replay_compression	random	novelty_gate	27	0.098726	0.0	0.062826	0.011577	0.817438	0.76	0.262174	0.16

Condition	Replay Mode	Selection Mode	Archive Size	Final TSP LIB Gap	Final Synthetic Gap	Final Transfer Gap	Adaptation Efficiency	Code Novelty (Mean)	Complexity	Failure Concentration (Mean)	Replay Failure Concentration (Final)
phase6_stratified_random_replay	stratified_random	score_only	29	0.121156	0.010343	0.08086	0.0	0.0	0.76	0.0	0.251701
phase6_diversity_weighted_replay	diversity_weighted	score_only	30	0.13595	0.0	0.086514	0.023865	0.827986	0.76	0.261003	0.130039
phase6_residual_failure_replay	residual_failure	score_only	29	0.082341	0.0	0.052399	0.039093	0.832596	0.76	0.148354	0.33564
phase6_diversity_failure_replay	diversity_failure	score_only	30	0.096401	0.0	0.061346	0.032532	0.828815	0.76	0.199284	0.17024
phase6_diversity_failure_replay_compression	diversity_failure	novelty_gate	29	0.234344	0.022049	0.157146	0.0	0.0	0.76	0.201519	0.188209

- **Best synthetic (lowest synthetic gap):** phase6_random_replay and its compression variant (0.0 synthetic gap).
- **Best held-out TSPLIB gap:** phase6_residual_failure_replay (0.082341), followed by phase6_random_replay_compression (0.098726).
- **Best transfer gap:** phase6_residual_failure_replay (0.052399), followed by phase6_random_replay_compression (0.062826).
- **Best overall transfer performance:** phase6_residual_failure_replay.
- **phase6_no_replay** has worst transfer and heldout TSPLIB gap, despite high code novelty.

Transfer Analysis and Replay Mechanism Insights

- **phase6_residual_failure_replay** (best transfer):
 - Replay Mode: residual_failure
 - Low heldout TSPLIB gap (0.0823) and best transfer gap (0.0524).
 - Archive hardness and failure concentration fairly high (~0.3), indicating focused replay on hard residual failures.
 - Moderate archive descriptor diversity (0.15).
 - Code novelty remains high (~0.83 mean).
 - Indicates that replay focused on residual failures yields best transfer to held-out TSPLIB.
- **phase6_random_replay** (best synthetic):
 - Replay Mode: random
 - Zero synthetic gap, good heldout TSPLIB gap (0.122) and transfer gap (0.078).

- Replay failure concentration moderate (~0.19 final).
- Archive diversity high (~0.24), size bias zero.
- Moderate code novelty (~0.69 mean).
- Shows broad coverage random replay supports synthetic performance but less so on real TSPLIB.
- **phase6_failure_replay**:
- Replay Mode: failure
- Higher heldout TSPLIB gap (0.213) and transfer gap (0.159), worse than random replay.
- Highest failure concentration (~0.26-0.31), low behavior profile complexity (0.56).
- Somewhat elevated archive hardness (~0.35) and size bias (0.12).
- Zero compression pressure but lower code novelty (~0.79), no transfer improvement.
- Replay on all failures without residual focus less effective.
- **Compression-aware replays** (random_replay_compression and diversity_failure_replay_compression):
- Showed reduced final TSPLIB gap and transfer gap compared to their non-compression counterparts.
- However, compression correlated with drops in code novelty (0.0 in some cases) without consistent large transfer gains.
- Indicating memory pressure may reduce lexical novel exploration but not necessarily harm transfer.
- **phase6_diversity_weighted_replay** and **phase6_diversity_failure_replay**:
- Intermediate transfer gaps (0.061-0.086).
- Replay failure concentration moderate (0.13-0.26).
- Maintain high code novelty (0.82+).
- Behaviors cluster around "clustered_constructor" or "balanced".
- Indicates diversity-weighted replay effective at balancing coverage and failure focus.
- **phase6_no_replay**:
- Worst TSPLIB gap (0.232), highest transfer gap (0.155).
- No replay results in minimal replay failure concentration (0.0).
- Highest code novelty by a margin (0.805).
- Suggests code novelty alone without replay does not improve transfer.

Mechanism Studies Summary

- **Archive Diversity**:
- Broad coverage replay (random, stratified_random, diversity_weighted) maintain high archive diversity (~0.24).
- Failure and residual failure replays have lower diversity (0.13-0.17).
- Diversity-weighted replay balances high diversity plus targeted failure replay.

- **Archive Hardness:**
 - Failure and residual failure playback achieve highest hardness (~0.3-0.35).
 - Random replay and no replay conditions have lower hardness (~0.15-0.18).
 - Hardness indicates replay targets more challenging failure instances.
- **Archive Size Bias:**
 - Failure-related archive keys (worst_archive) show size bias (~0.12), indicating replay preferences for larger instances.
 - Experience archive-based replays have zero size bias.
- **Failure Concentration:**
 - Failure and residual failure replay conditions concentrate replay on failure modes (0.17-0.33 final).
 - Random replay less so (~0.16-0.22).
 - No replay zero by definition.
- **Failure Residuals (mean_selected_residual_gap):**
 - Residual failure replay achieves negative mean residual gaps consistently (-0.13 to -0.27), suggesting targeted reduction of systematic failures.
 - Other replays show mixed or near-zero residual gaps.

Code Novelty vs Transfer Tradeoff

- Highest code novelty in no replay (0.805) but worst transfer and TSPLIB gaps.
- Residual_failure_replay achieves best transfer but slightly reduced code novelty (~0.83) - still high.
- Random replay variants show slight decrease code novelty in exchange for improved transfer.
- Compression pressure reduces code novelty to zero in novelty_gate selection modes but may improve heldout TSPLIB and synthetic gaps modestly.
- Overall, code novelty and transfer are related but not strictly correlated; replay focused on hard residuals trades some novelty for better transfer.

Summary of Replay Condition Categories

- **Broad-coverage replay:** random, stratified_random, diversity_weighted
 - High archive descriptor diversity (~0.24)
 - Moderate hardness (~0.15-0.19)
 - Moderate failure concentration (~0.13-0.26)
 - Good synthetic holdout performance (0 or near-zero gaps)
 - Moderate transfer performance (TSPLIB gaps approx 0.1-0.14)
- **Raw-failure replay:** failure replay

- High hardness (~0.35)
- High final failure concentration (~0.31)
- Largest archive size bias (~0.12)
- Higher TSPLIB gaps (about 0.21) and higher transfer gaps (about 0.16)
- Lower code novelty (~0.79)
- **Residual-failure replay:** residual_failure replay
 - High hardness (~0.34)
 - Moderate diversity (~0.15)
 - Moderate failure concentration (~0.17)
 - Best TSPLIB and transfer gaps (0.082 and 0.052)
 - High code novelty (~0.83)
- **Diversity-weighted failure replay:** diversity_failure replay
 - Archive hardness about 0.29-0.32
 - Archive diversity moderate (0.17)
 - Failure concentration ~0.17-0.20
 - TSPLIB gaps around 0.09-0.1
 - Transfer gaps around 0.06
 - High code novelty (~0.83)
- **Compression-aware replay:** random_replay_compression and diversity_failure_replay_compression
 - Compression reduces code novelty to 0.0 in novelty_gate selection mode
 - Slight improvements in TSPLIB gaps and transfer gaps compared to non-compression variants
 - Archive diversity remains high

Conservative Interpretation

1. **Best Transfer:**

- phase6_residual_failure_replay achieves the best TSPLIB and transfer gaps, showing that replay focused on residual failures improves transfer generalization.
- High archive hardness and moderate diversity supports efficient replay of challenging failures.

2. **Synthetic Holdout:**

- phase6_random_replay reaches zero mean synthetic holdout gap, suggesting broad coverage replay is sufficient for synthetic test cases.
- This comes with moderate transfer and TSPLIB improvements but less than residual_failure replay.

3. **No Replay:**

- Highest code novelty but worst gaps and no replay failures or failure concentration.

- Code novelty alone does not translate into transfer gain.

4. **Failure Replay (raw failures):**

- High failure concentration and hardness but transfer gaps remain high.
- Suggests replaying raw failures without residual focus may not efficiently drive transfer.

5. **Compression Effects:**

- Compression reduces code novelty sharply in novelty_gate selections but can modestly improve transfer metrics.
- Shows a tradeoff between code novelty and replay efficiency under memory pressure.

6. **Diversity-weighted and diversity failure replay:**

- Offer a balanced compromise, good code novelty and transfer performance with moderate failure concentration.

Recommendations

- Prioritize **residual failure replay** mechanisms for improved transfer performance on TSPLIB benchmarks.
- Incorporate **broad coverage replay** (random or diversity-weighted) to maintain zero synthetic holdout gaps.
- Use compression carefully; while it may improve transfer, it reduces code novelty substantially.
- Avoid relying solely on code novelty without replay, as it does not yield transfer benefits.
- Consider failure concentration and archive hardness as key replay archive properties influencing transfer.
- Distinguish replay modes carefully as they have systematic effects on archive properties and generalization.

This conservative analysis focuses on quantitative gaps and replay archive metrics to inform transfer improvements in replay-aware TSP benchmarks.